

Trajectory Forecasting for Autonomous Vehicle on Highway: A Graph Neural Network based Approach

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Abstract—Trajectory forecasting is the most important task in motion planning in autonomous vehicles. Despite technological advancements, self-driving remains challenging. Currently, the availability of big datasets and advanced machine learning models plays a crucial role to capture and understand the complex behavior of real-world road situations. In this work, an encoder-decoder based deep learning model GAT-GRU is proposed to capture spatio-temporal interaction between vehicles on the highway by using real-world dataset NGSIM US-101. This model captures interaction dynamics between the vehicles through graphs and temporal dependencies for predicting the future trajectory of an ego vehicle. This approach achieves modest accuracy in short-term trajectory prediction; it reduces average displacement error (ADE) and final displacement error (FDE) by more than 69% and 21% respectively over baseline approaches. These gains demonstrate the value of integrating both spatial attention and temporal memory for accurate, robust trajectory forecasting in highway scenarios, showing its effectiveness in real-world highway driving.

Index Terms—Graph Attention Network(GAT), Gated Recurrent Unit(GRU), Interaction Aware, Trajectory Prediction, Autonomous Vehicles

I. INTRODUCTION

Trajectory forecasting in an accurate manner is essential for autonomous driving systems, it allows neighboring vehicles to anticipate the behavior and plan safely for efficient maneuvers. In highway driving scenarios, where vehicles frequently change lanes, accelerate, decelerate, and interact with each other, predicting future position in that kind of scenario becomes particularly challenging because of the dynamic and interactive nature of multi-agent interactive environments.

Earlier solution of such a complex problems relied on physics-based models that assume constant velocity or acceleration, such as Kalman Filters or constant-turn-rate models. Although these models are computationally

efficient, but these approaches are limited in their ability to adapt to complex interactions and human-like behaviors or contextual traffic interactions. The utilization of models driven by data for trajectory forecasting has been made possible due to recent developments in deep learning. In particular, temporal dependencies in motion sequences have been effectively modeled by recurrent neural networks (RNNs). The Social Long Short-Term Memory (LSTM) in [1] proposed by Alahi et al. to model social interactions by using pooled hidden states of neighboring agents in crowded pedestrian environments. However, these pooling mechanisms are often hand-engineered and may fail to capture the varying influence of different agents. Researchers have moved to graph-based models to deal with this, which explicitly encode agent relationships using Graph Neural Networks (GNNs). In order to allow the model to focus on more significant neighbors, in [2] Vemula et al. presented a GAT that learns attention weights between agents. This was further extended in works such as Trajectron++ [3] and ST-GAT [4], which demonstrated that graph-structured spatio-temporal modeling can significantly improve trajectory forecasting accuracy in multi-agent environments.

This work proposes a graph-based GAT-GRU encoder-decoder framework for trajectory prediction on the NGSIM US-101 highway dataset. Proposed method models each scene as a time series of ego-centric graph interactions, graphs at each time step by selecting vehicles within a distance threshold, where each node represents a vehicle and edges represent spatial proximity-based interactions. Node features include position, velocity, acceleration, and lane ID, while the spatial attention mechanism enables the model to learn which vehicles are most influential in each scene. The GRU-based encoder-

decoder architecture then captures temporal dynamics and forecasts the ego vehicle's future trajectory.

Contributions:

- Proposed a complete pipeline for constructing ego-centric, spatially normalized interaction graphs from the NGSIM highway dataset.
- Introduced a GAT-GRU encoder-decoder model that captures both spatial and temporal dependencies in multi-vehicular highway scenarios.
- Evaluated proposed model on real highway driving data and demonstrated modest performance in short-term trajectory forecasting, using average and final displacement errors as evaluation metrics.

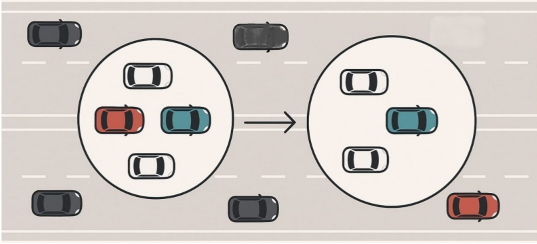


Fig. 1. Overview of Highway Scenario to capture spatial interaction by Graph construction.

II. RELATED WORK

A. Trajectory Prediction

In early research on trajectory forecasting of vehicles, mostly depend on physics-based models. The Kalman filter method considers noises within the construction of model and predicts vehicle trajectories through Gaussian distribution modeling [6]. But Kalman filter is used only for linear scenarios, on the other hand, Monte Carlo method provides benefit to handle non linear and multi-modal scenarios [7].

Data-driven approaches have taken the forefront with the advent of deep learning. RNNs, especially LSTM networks, are widely used for capturing temporal dependencies in motion sequences. Using a social tensor and a spatial grid to capture dynamic vehicle characteristics, Deo and Trivedi [8] processed the data using a 2D Convolutional Neural Network (CNN) for spatial dependencies and an LSTM for temporal encoding. Though effective, such pooling mechanisms lack flexibility in capturing heterogeneous interactions.

B. Graph-Based Interaction Models

Recent work has shown that GNNs are highly effective for modeling inter-agent interactions. In [12], Zihao Sheng et al. proposed a spatio-temporal model using Graph Convolutional Network (GCN) for spatial encoding and CNN for temporal features. Unlike GCNs, which aggregate neighbor information uniformly

or based on degree normalization, GATs enable the model to concentrate on the most influential agents, improving interpretability and forecasting accuracy in complex, dense traffic scenes. In [2], Vemula et al. introduced a GAT for social behavior prediction, enabling the model to learn the importance of different neighboring agents through attention weights. This idea has been successfully adapted to vehicular trajectory prediction, where interaction graphs model vehicles as nodes and spatial relationships as edges.

Ye and Chen proposed GTNet, a graph-transformer-based model that captures the interactions locally as well as globally in a traffic scene for New Energy Vehicles (NEVs) [9]. In a related direction, Gao et al. used a GAT-GRU encoder-decoder framework to predict the trajectories of surrounding vehicles under mixed traffic conditions, demonstrating improved displacement error performance compared to GRU-only baselines [10].

The NGSIM US-101 dataset has been extensively used for trajectory modeling due to its high-resolution tracking of real-world traffic on highways. Several works have used it for lane change prediction [5], trajectory generation, and behavior classification. In [11], K. Zhang et al. presented an attention-based interaction-aware trajectory forecasting framework called AI-TP, by utilizing GAT and ConvGRU, which achieved superior accuracy and faster inference on datasets like NGSIM, ApolloScape, and Argoverse.

III. PROBLEM DEFINITION

Given a spatio-temporal dataset of vehicle trajectories collected from highway traffic scenes, the ultimate objective is to forecast the future coordinates of a designated ego vehicle over a fixed prediction horizon.

Formally, for each time step t , assume that a set of vehicles is present in the scene. Each vehicle i is associated with a feature vector $x_t^i \in \mathbb{R}^F$, where $F = 5$ (representing position, velocity, acceleration, and lane ID). A dynamic interaction graph $G_t = (V_t, E_t)$ is constructed to represent agent relationships based on spatial proximity.

Given a sequence of graphs $\{G_{t-T+1}, \dots, G_t\}$, the goal is to predict the ego vehicle's future positions for the upcoming H time steps:

$$\hat{y} = \{\hat{y}_{t+1}, \dots, \hat{y}_{t+H}\}, \quad \hat{y}_{t+h} \in \mathbb{R}^2$$

where each $\hat{y}_{t+h}$ represents the predicted (x, y) location of the ego vehicle.

The model must learn a mapping:

$$f : \{G_{t-T+1}, \dots, G_t\} \rightarrow \{\hat{y}_{t+1}, \dots, \hat{y}_{t+H}\}$$

IV. DATA SOURCE AND DESCRIPTION

The data used in this work is derived from the Next Generation Simulation (NGSIM) US-101 dataset [13], released by the U.S. Federal Highway Administration. The dataset provides high-resolution (10 Hz) vehicle trajectory recordings over a 640-meter road segment of the US-101 highway in Los Angeles in the state of California. The recordings span approximately 45 minutes and include hundreds of vehicles moving in a dense multi-lane freeway environment.

Each row in the dataset corresponds to a single vehicle at a particular time frame and contains the following relevant attributes:

- Vehicle ID: Unique identifier for each vehicle
- Frame ID: Time step index (recorded at 10 Hz)
- Local X / Local Y: Position coordinates of the vehicles (in feet) relative to a fixed origin
- Vel: Instantaneous speed (ft/s)
- Acc: Instantaneous acceleration (ft/s²)
- Lane ID: The current lane number in integer
- Global Time: Timestamp in milliseconds
- Additional features such as lane change indicators, vehicle dimensions, and preceding/following IDs etc.

V. METHODOLOGY

This section outlines a proposed encoder-decoder architecture in Fig. 2 that uses a series of graph-structured traffic data to learn spatio-temporal interactions to forecast the trajectory.

A. Graph-Based Modeling

The proposed trajectory forecasting model integrates spatial encoding via GAT, social pooling, a temporal GRU encoder-decoder, encoder-decoder attention, and a composite loss function.

- *Encoder*: Each traffic graph is encoded using a GAT to model spatial interactions, followed by social pooling to capture neighboring context. The resulting spatio-social features are processed by a GRU to encode temporal dynamics.
- *Decoder*: A GRU-based decoder, initialized with the encoder's final state, predicts future positions autoregressively. At each step, it attends to encoder outputs via encoder-decoder attention and uses the previous output (or ground truth) with the context to generate the next position.

B. Spatial Encoding with Graph Attention Network

A graph $G_t = (V_t, E_t)$ represents the scene for every time steps t , where V_t denotes the set of vehicles and E_t their spatial interactions. Each node $i \in V_t$ has features

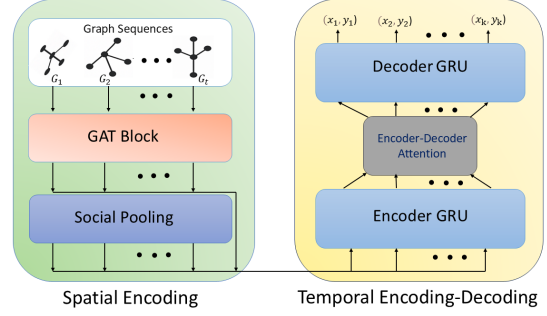


Fig. 2. Proposed GAT-GRU model.

$\mathbf{x}_i^t \in \mathbb{R}^F$. The GAT computes attention-based spatial embeddings as:

$$e_{ij} = \text{LeakyReLU}(\mathbf{a}^\top [\mathbf{W}\mathbf{x}_i \parallel \mathbf{W}\mathbf{x}_j]), \quad (1)$$

$$\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_{k \in \mathcal{N}(i)} \exp(e_{ik})}, \quad (2)$$

$$\mathbf{h}_i = \sum_{j \in \mathcal{N}(i)} \alpha_{ij} \mathbf{W}\mathbf{x}_j, \quad (3)$$

where $\mathbf{W} \in \mathbb{R}^{d \times F}$ and $\mathbf{a} \in \mathbb{R}^{2d}$ are learnable parameters, and node i 's neighbors are denoted by $\mathcal{N}(i)$. The embedding for the ego vehicle $\mathbf{h}_{\text{ego}}$ is extracted for further processing.

C. Social Pooling

To model social interactions, applied a learned pooling function $\phi(\cdot)$ over the neighbors of the ego vehicle:

$$\mathbf{z}_{\text{pool}} = \frac{1}{|\mathcal{N}_{\text{ego}}|} \sum_{j \in \mathcal{N}_{\text{ego}}} \phi(\mathbf{h}_j), \quad (4)$$

$$\phi(\cdot) = \text{ReLU}(\mathbf{W}_2 \text{ReLU}(\mathbf{W}_1 \cdot)), \quad (5)$$

and concatenate this with the ego embedding:

$$\mathbf{z}_t = [\mathbf{h}_{\text{ego}} \parallel \mathbf{z}_{\text{pool}}] \in \mathbb{R}^{2d}. \quad (6)$$

D. Temporal Encoding with GRU

The temporal dynamics of the ego vehicle are encoded using a GRU over the sequence $\{\mathbf{z}_{t-T+1}, \dots, \mathbf{z}_t\}$:

$$\mathbf{h}_t^{\text{enc}} = \text{GRU}(\mathbf{z}_t, \mathbf{h}_{t-1}^{\text{enc}}), \quad (7)$$

where the final hidden state $\mathbf{h}_T^{\text{enc}}$ summarizes the trajectory history.

E. Encoder-Decoder Attention

To allow the decoder to attend to relevant past states, used an encoder-decoder attention mechanism. At each decoder step h , a context vector $\mathbf{c}_h$ is computed as:

$$e_{h,t} = \frac{\mathbf{q}_h^\top \mathbf{h}_t^{\text{enc}}}{\sqrt{d}}, \quad (8)$$

$$\alpha_{h,t} = \frac{\exp(e_{h,t})}{\sum_{t'=1}^T \exp(e_{h,t'})}, \quad (9)$$

$$\mathbf{c}_h = \sum_{t=1}^T \alpha_{h,t} \mathbf{h}_t^{\text{enc}}, \quad (10)$$

where $\mathbf{q}_h$ is the decoder query vector, typically the previous decoder hidden state.

F. Decoder with Teacher Forcing

The decoder GRU is initialized with $\mathbf{h}_T^{\text{enc}}$ and runs autoregressively. At each future step $h \in \{1, \dots, H\}$:

$$\mathbf{x}_h^{\text{dec}} = [\hat{\mathbf{y}}_{t+h-1} \parallel \mathbf{c}_h], \quad (11)$$

$$\mathbf{h}_{t+h}^{\text{dec}} = \text{GRU}(\mathbf{x}_h^{\text{dec}}, \mathbf{h}_{t+h-1}^{\text{dec}}), \quad (12)$$

$$\hat{\mathbf{y}}_{t+h} = \mathbf{W}_o \mathbf{h}_{t+h}^{\text{dec}} + \mathbf{b}_o, \quad (13)$$

where $\hat{\mathbf{y}}_{t+h} \in \mathbb{R}^2$ is the predicted position. During training, ground truth positions are used in place of predictions with a probability p (teacher forcing).

G. Loss Function

Mean Squared Error (MSE) and Huber loss are used to train the model:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{H} \sum_{h=1}^H \|\hat{\mathbf{y}}_{t+h} - \mathbf{y}_{t+h}\|_2^2, \quad (14)$$

$$\mathcal{L}_{\text{Huber}} = \frac{1}{H} \sum_{h=1}^H \text{Huber}(\hat{\mathbf{y}}_{t+h}, \mathbf{y}_{t+h}) \quad (15)$$

During evaluation, ADE and FDE is reported.

$$\text{ADE} = \frac{1}{H} \sum_{h=1}^H \|\hat{\mathbf{y}}_{t+h} - \mathbf{y}_{t+h}\|_2 \quad (16)$$

$$\text{FDE} = \|\hat{\mathbf{y}}_{t+H} - \mathbf{y}_{t+H}\|_2 \quad (17)$$

These metrics provide intuitive measures of how the predicted trajectory resembles the actual future path.

VI. EXPERIMENTS

A. Experimental Setup

1) *Data Preprocessing and Graph Construction:* The raw NGSIM US-101 dataset was originally in CSV format and was converted to graph-structured data suitable for learning of the model. The pseudocode for graph data construction is in Algorithm 1. The direction of a vehicle is predetermined for each ego vehicle to forecast the future coordinates in a deterministic manner on a highway setting. A graph sequence of five past frames to encode the spatio-temporal history of the traffic scene and four future frames for the prediction horizon is used.

2) Details of Implementation:

Libraries: For graph creation NetworkX and model implementation PyTorch Geometric and PyTorch have been used.

Dataset Split: The dataset has been divided as follows: 80% has been used for training, while the remaining 20% has been used for validation. However, to avoid driver-specific leakage across training and test sets, adopted a scene-level split in line with prior works such as Trajectron++ and Wayformer.

Normalization:

- Positions and future coordinates are normalized relative to the ego vehicle.
- Velocities and accelerations are scaled to standard ranges.
- Lane IDs are normalized to the $[0, 1]$ range.

Training Details:

The model was trained using the following settings:

- Batch Size: 1, due to training and prediction for each ego vehicles.
- Hardware: NVIDIA K80 GPU
- Training Time: Approximately 0.5 hours for 30 epochs over the full dataset

To support high-traffic highway scenarios, this model processes each frame as an ego-centric graph that includes nearby vehicles within a fixed radius. This design naturally adapts to varying traffic densities. The GAT layer helps the model focus on the most influential agents, even in crowded scenes. During training, it is also ensure that the model sees a wide range of traffic conditions from sparse to very dense, so it learns to handle real-world highway behaviour.

B. Model Hyperparameters

Parameter	Value
GAT hidden dimension	64
GRU hidden dimension	64
Learning rate	1×10^{-3}
Epochs	20, 30
Optimizer	Adam
Dropout	0.3, 0.4
Weight initialization	Random, Xavier
Weight Decay	1×10^{-5}
Teacher forcing	0.5
Loss Function	MSE, Huber Loss

C. Evaluation Metrics

ADE and FDE are utilized to evaluate the proposed trajectory forecasting model's performance. ADE calculates the average variations between the predicted and actual positions over the whole prediction horizon, providing an overall indication of trajectory accuracy. FDE focuses on the distance at the final predicted time step, reflecting the model's ability to estimate the long-term position accurately.

Algorithm 1 Ego-Centric Graph Construction

```

1: Input: Dataset  $\mathcal{D}$ , dist threshold  $\tau$ , max nodes  $N$ ,
   graph steps  $T_g$ , future steps  $T_f$ 
2: Output: Feature sequences  $F$ , adjacency matrices
    $A$ , future coordinates  $Y$ 
3: Initialize  $F \leftarrow []$ ,  $A \leftarrow []$ ,  $Y \leftarrow []$ 
4: for each unique vehicle ID  $v_e$  in  $\mathcal{D}$  do
5:   Extract  $\mathcal{D}_{v_e} \leftarrow \{x \in \mathcal{D} \mid x.\text{Vehicle\_ID} = v_e\}$ 
6:   for each direction  $d$  in  $\mathcal{D}_{v_e}$  do
7:     Filter  $\mathcal{D}_{v_e}^d \leftarrow \{x \in \mathcal{D}_{v_e} \mid x.\text{Direction} = d\}$ 
8:     Sort  $\mathcal{D}_{v_e}^d$  by Global_Time ascending
9:     Extract frame list  $T \leftarrow [f_1, f_2, \dots, f_n]$ 
10:    Initialize  $graph\_count \leftarrow 0$ ,
        $future\_count \leftarrow 0$ 
11:    for each frame ID  $f_t$  in  $T$  do
12:       $frame\_data \leftarrow \{x \in \mathcal{D} \mid$ 
        $x.\text{Frame\_ID} = f_t \wedge x.\text{Direction} = d\}$ 
13:      if  $graph\_count < T_g$  then
14:        if  $v_e \in frame\_data$  then
15:          Extract ego position  $(x_e, y_e)$ , Cre-
          ate graph  $G$  with node  $v_e$ 
16:          for each vehicle  $v_j \neq v_e$  in
              $frame\_data$  do
17:            if  $\text{distance}(v_e, v_j) < \tau$  then
18:              Add node  $v_j$  and edge
               $(v_e, v_j)$  to  $G$ 
19:            end if
20:          end for
21:          Append features and adjacency
          matrix of  $G$  to  $F$ ,  $A$ 
22:          Update  $last\_graph \leftarrow G$ ,
              $last\_position \leftarrow (x_e, y_e)$ 
23:          else
24:            Append  $last\_graph$ 's features
            and adjacency to  $F$ ,  $A$ 
25:          end if
26:          Increment  $graph\_count$ 
27:        else
28:           $future\_count < T_f$ 
29:          if  $v_e \in frame\_data$  then
30:            Extract  $(x_e, y_e)$ 
31:            Append to  $Y$ 
32:            Update  $last\_position \leftarrow (x_e, y_e)$ 
33:          else
34:            Append  $last\_position$  to  $Y$ 
35:          end if
36:          Increment  $future\_count$ 
37:        end if
38:      end for
39:    end for
40:  end for
41: return  $F, A, Y$ 

```

VII. QUANTITATIVE RESULTS & ANALYSIS

Comparison of proposed **GAT-GRU** model with base-
line models:

- *Constant Velocity (CV)*: A naive baseline that extrapolates future positions assuming constant speed and direction, resulting in high prediction errors.
- *Simple GRU*: A sequence model that uses only past ego trajectories, without incorporating spatial interactions, offers moderate improvement over CV.
- *MLP*: A feedforward network trained on past trajectory coordinates, capturing some temporal patterns but lacking sequential dynamics.
- *GAT-GRU*: The proposed model integrates spatial (via GAT) and temporal (via GRU) reasoning, achieving the best performance across all metrics.

TABLE I
PERFORMANCE COMPARISON OF TRAJECTORY PREDICTION
MODELS FOR MSE TRAINING LOSS.

Model	ADE (ft / m)	FDE (ft / m)
Constant Velocity	2738.98 / 834.69	4692.50 / 1430.25
Simple GRU	1077.08 / 328.23	450.35 / 137.27
MLP	1023.84 / 312.11	704.57 / 214.75
GAT-GRU	312.35 / 95.21	351.60 / 107.20

The GAT-GRU model achieves the lowest ADE of 312.35 ft and FDE of 351.60 ft, which significantly outperforms all baselines. Compared to the CV model, GAT-GRU reduces ADE and FDE by approximately 88.6% and 92.5%, respectively. Also, against the Simple GRU baseline, GAT-GRU achieves a reduction of 71.0% in ADE and 21.9% in FDE. Relative to the MLP model, the improvement is 69.5% in ADE and 50.1% in FDE. These results highlighted the advantages of incorporating both spatial (via GAT) and temporal (via GRU) context in modeling agent trajectories.

This work presents a baseline GAT-GRU model with considerations from the ego-centric proximity-based graph creation to modeling. This model focuses on highway scenarios where vehicles are moving at a fast pace to forecast the future using the NGSIM US-101 dataset, whereas many state-of-the-art (SOTA) models emphasize dense urban environments, often including rich map semantic information and trained on datasets such as nuScenes, Argoverse, or ETH/UCY. A direct numerical contrast with urban-focused SOTA methods is meaningless because of these variations in the scenario, data organization, and modeling objectives. Rather, using the same dataset and configuration, the proposed model was compared against classical baselines. In future studies, this model can be expanded to more comprehensive benchmarks.

VIII. ABLATION STUDY

An ablation study was conducted by altering essential components and observing their impacts on performance in order to assess the contributions of various parts in this model architecture. The proposed **GAT-GRU** model is compared with two models:

- *GAT-GRU (baseline)*: This combines a Graph Attention Network with a GRU encoder-decoder for temporal and relational trajectory modeling.
- *GAT-GRU with Social Pooling*: This improves the baseline by spatially pooling around vehicles to incorporate social context.
- *GAT-GRU with Social Pooling and Encoder-Decoder Attention*: Adds attention between encoder and decoder states with social pooling to capture accurately temporal dependencies and critical interactions.

TABLE II
PERFORMANCE COMPARISON OF TRAJECTORY PREDICTION
MODELS FOR MSE TRAINING LOSS.

GAT-GRU Model	ADE (ft / m)	FDE (ft / m)
Baseline	312.35 / 95.22	351.60 / 107.20
Socialpooling	340.87 / 103.88	364.79 / 111.16
Socialpooling Attention	367.71 / 112.05	368.18 / 112.22

TABLE III
PERFORMANCE COMPARISON OF TRAJECTORY PREDICTION
MODELS FOR HUBER TRAINING LOSS.

GAT-GRU Model	ADE (ft / m)	FDE (ft / m)
Baseline	297.74 / 90.74	346.47 / 105.58
Socialpooling	358.58 / 109.31	378.77 / 115.42
Socialpooling Attention	395.58 / 120.56	405.27 / 123.52

Evaluated three versions of trajectory forecasting model under two training regimes: using Mean Squared Error (MSE) and Huber loss.

1) *Statistical Significance*: Performance comparison of trajectory prediction models using MSE loss. Values shown as mean $\pm$ std (in feet). $\dagger$ indicates statistical significance over GAT-GRU ($p < 0.05$).

TABLE IV
TEST RESULTS SUMMARY (AVERAGED OVER 3 RUNS).

GAT-GRU Model	ADE (ft)	FDE (ft)	Significance
Baseline	312.33 $\pm$ 1.52	351.60 $\pm$ 2.58	–
SocialPooling	340.87 $\pm$ 4.16	364.80 $\pm$ 4.14	$\checkmark (p < 0.05)$
SocialPooling Attn.	368.63 $\pm$ 6.18	390.27 $\pm$ 7.58	$\checkmark (p < 0.05)$

$\dagger$ Indicates statistically significant performance degradation compared to the GAT-GRU baseline ($p < 0.05$).

This result supports the observation that added architectural complexity, such as attention layers, does not necessarily improve performance, may harm generalization due to overfitting or insufficient data coverage.

IX. DISCUSSION

The proposed GAT-GRU model demonstrates modest performance in forecasting vehicle trajectories. There are several limitations and challenges worth discussing. First, the current formulation assumes the ego vehicle is always the focal point and that interactions are only unidirectional (i.e., from neighbors to ego). More advanced models could capture bidirectional influences or fully connected interactions. Second, the graph construction is based on a fixed spatial threshold and a fixed number of nodes, which may limit scalability in highly dense traffic. Also, attention-based edge weighting could improve flexibility. Third, the model currently predicts a single deterministic trajectory. But, real-world driving involves inherent uncertainty and multimodal behaviors (e.g., lane changes or yielding). Finally, the use of hand-crafted normalization and fixed time windows limits adaptability in varying scenarios. An end-to-end trainable preprocessing pipeline or reinforcement learning integration could help the model learn scene-specific dynamics more effectively.

X. CONCLUSIONS

A graph-based spatio-temporal trajectory forecasting framework has been presented in this paper. It uses a GAT-GRU encoder-decoder model. This approach models traffic scenes as ego-centric graph sequences. Experiments on the NGSIM US-101 highway dataset demonstrated that this approach significantly reduces ADE and FDE compared to baseline methods.

Here, forecasts are restricted to short-term predictions (0.4s), which may not suffice for long-term planning. The model is validated only on highway data, without testing in urban or diverse environments. Future work may explore multimodal prediction or flow-based methods to better handle uncertainty, incorporate edge features, a distance threshold for graph construction as a hyperparameter, and a maximum number of vehicles under the distance threshold of the ego vehicle that are allowed. It is also necessary to extend the forecast horizon to at least two seconds by using past five-second data to support downstream decision making, and test on diverse datasets like Waymo or nuScenes in complex traffic scenes.

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